

GE 502 : Consciousness and Professional Ethics

Course Code : GE 502

Course Name : Consciousness and Professional Ethics

L-T-P-C : 2-1-0-3

Intended for : UG/PG/PhD

Prerequisite :

Mutual Exclusion:

Approval: 54th BoA

Course Contents

- **Introduction and Background:** Evolution of the engineering profession; Basis for universal human values and ethical human conduct; Engineering profession in the light of comprehensive human goal; Responsibility in engineering; Social and value dimensions of technology; Continuous happiness and prosperity as basic human goal?; connection of basic human goal with professional conduct; process of exploring human goal; Summary of few cases of professional ethics code developed by professional societies. (4 Hours)
- **Exploring Harmony in the Human Being and Consciousness:** Understanding human being as a co-existence of the sentient consciousness and the material 'Body', Understanding the needs of consciousness and body - happiness and physical facility, Appraisal of Physical needs, meaning of Prosperity in detail. (4 Hours)
- **Harmony in the Family (Human-Human Interaction):** Understanding values in humanhuman relationship; meaning of Justice and program for its fulfilment to ensure mutual happiness; Trust and Respect as the foundational values of relationship, Understanding the meaning of Trust; Difference between intention and competence, Understanding the meaning of Respect, Difference between respect and differentiation; the other salient values in relationship, Notion of Justice as fulfilling relationship. (4 Hours)
- **Harmony in the Society (Human-Human Interaction):** Understanding the harmony in the society (society being an extension of family): Resolution, Prosperity, fearlessness (trust) and co-existence as comprehensive Human Goals, Visualizing a universal harmonious order in societyUndivided Society. (4 Hours)
- **Understanding Harmony in the Nature and Existence:** Whole existence as Coexistence: Understanding the harmony in the Nature, Interconnectedness and mutual fulfillment among the various units of nature, recyclability and self-regulation in nature. Understanding Existence as Co-existence of mutually interacting units in all pervasive space, Holistic perception of harmony at all levels of existence. (4 Hours)
- **Implications of the Holistic Understanding of Harmony on Professional Ethics:** Natural acceptance of human values, Definitiveness of Ethical Human Conduct, Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order, Competence in professional ethics: (i) Ability to utilize the professional competence for augmenting universal human order (ii) Ability to identify the scope and characteristics of people friendly and ecofriendly production systems, (iii) Ability to identify and develop appropriate technologies and management patterns for above production systems. Case studies of typical holistic technologies, management models and production systems. (5 Hours)
- **Strategy for transition from the present state to Universal Human Order:**

1. At the level of individual: as socially and ecologically responsible engineers, technologists and managers
2. At the level of society: as mutually enriching institutions and organizations (3 Hours)

Laboratory/practical/tutorial Modules:

Lecture hours are to be used for interactive discussion, placing the proposals about the topics at hand and motivating students to reflect, explore and verify them for developing a holistic view so that basis for professional ethics can be established. Tutorial hours are to be used for practice sessions. While analyzing and discussing the topic, the instructor's role is in pointing to essential elements to help in sorting them out from the surface elements. In other words, help the students explore the important or critical elements. In the discussions, particularly during practice sessions (tutorials), the instructor encourages the student to connect with one's own self and do self-observation, self-reflection and self-exploration. Scenarios may be used to initiate discussion. Observations and their analyses are shared and discussed among all students, in a group sitting. Indicative plan of fourteen tutorials is as follows: Differentiating the notions of Preconditioning, Sensation and Natural Acceptance; Notions of Happiness; Case study of current economic activities and associated expectations of humans through the short video story of 'stuff'; Self-awareness through the list and analysis of individuals' desires; Self-awareness and individuals' role in ensuring the healthy condition of the body; Self-assessment of our status of prosperity; Appreciation of impact of individual behaviors on others through short movie 'Right Here Right Now'; Observing the feeling of gratitude; Discussion through short movie 'Economics of Happiness' for understanding the role in localization and globalization; Discussion through short movie 'Swaraj in Hiware Bazar' for appreciating the possibility of transformation through relationship; Discussion through short movie 'An inconvenient truth' for understanding the role of humans for mitigation of global warming; Discussion on the role of students for improving the status of justice in the institute and asking them to make a framework; Discussion on role of students for improving the economic sustainability of the institute; Discussion on role of students in reducing the carbon emissions impact of the institute; Differentiating the internal and external sources of happiness.

Textbooks:

1. R.R. Gaur, R. Asthana and G.P. Bagaria, **A Foundation Course in Human Values and Professional Ethics**, 3rd Revised Edition, UHV Publications, 2023.
2. W. R. Bowie, **Engineering Ethics: Outline of an Aspirational Approach**, Springer, 2009.
3. P. A. Vesilind, **Engineering Peace and Justice: The Responsibility of Engineers to Society**, Springer, 2010.

References:

1. C.E. Harris, M. S. Pritchard and M. J. Rabins, **Engineering Ethics: Concepts and Cases**, 4th Edition, Cengage Learning, 2009.
2. G. D. Baura, **Engineering Ethics: An Industrial Perspective**, Academic Press (Elsevier), 2006.
3. J. Lucena, **Engineering Education for Social Justice: Critical Explorations and Opportunities**, Springer, 2013.